Task 2: Ranking

For keyword matching, each candidate document, the score is 1 if the query and document share at least one token, otherwise 0. For the hybrid matching, scoring works by using cosine similarity between a query vector and document vectors in a TF-IDF matrix. Rarer terms get higher weights. The TF mode is raw frequency length and uses the IDF formula typically used for TF-IDF. The hybrid ranker uses three signals; are independently scored, transformed, normalised then combined. The TF-IDF raw frequency captures the lexical exact-matching. The TF-IDF log mode captures dampened term frequencies. And the dense-vector TF-IDF weighted captures semantic similarity such as synonyms. Each raw score is power transformed to the signal's power (2, 2, 4). (Higher powers amplify differences between high-scoring documents.) Then the scores are min-max scaled to 0 – 1 per query, so signals are comparable before combining. Weights are tuned for maximising average Pearson score.

	Keyword matching (no setting)	**Default (Hybrid)**
Average Pearson (AP) score	0.000	0.595

For query Q132 - "theoretical studies of creep buckling" keyword matching scores 1 for document 1014 as it shares "studies", "creep" and "buckling" with relevance 4. However, it also scores 1 for document 112 as it shares the word "studies", however its relevance is 0. Whereas the hybrid matching scores a higher score in the range [0, 1] for document 1014 than 112, correlating more with relevance. The TF-IDF signals mean that "theoretical" appears in 1014 but not in 112, giving 1014 a higher score. Also, the dense vector means that "theoretical studies" is semantically closer to the query than just for example "buckling" in isolation. This produces a vector where gold=4 documents consistently receive higher predicted scores than for example gold=2 documents, which roughly follows the linear relationship with gold grades. This is why keyword matching is not the final choice, as the covariance with gold is near 0 because the score does not track which documents are more relevant. Pearson returns 0. For query Q218 - "what is the heat transfer to a blunt body in the absence of vorticity" pure TF-IDF suggests that it cannot distinguish document 1274 (gold=0, mentions "blunt bodies" in a gas transfer context) from 559 (gold=2, context surrounding heat transfer to blunt bodies) because both share the same terms with similar frequencies. Hybrid's dense vector score separates different contexts where pure TF-IDF struggles.

Task 3: System Optimisation

The expanded query optimisation uses the WordNet corpus to expand the query terms using noun synonyms. This optimisation takes place in line 224 in search_system.py just before ranking for each query. This change is isolated from the preprocessing optimisation. The preprocessing optimisation includes removing stop words. As well as splitting hyphenated terms. This optimisation takes place in lines 201 and 214 for preprocessing both documents and queries for more term matching.

	No optimisation	**Preprocessing**	**Query Expansion**
Mean Average Pearson (MAP) score	0.2365	0.2801	0.2167

Under no optimisation a typical document has ~40 stop words, (high DF -> low IDF), so they do not help differentiation, diluting contribution of content terms. For query N09 "criterion for true panel flutter, small amplitude vibration, acoustic disturbances", it ranks document 319 as 1 which is irrelevant (contains "for" 3 times), whereas the preprocessing optimisation ranks document 658 as 1 which is relevant. The AP for N09 jumps from 0.146 to 0.613 with preprocessing, letting the TF-IDF signals properly rank documents that mention "panel", "flutter" prominently. This means that the MAP would also increase, as MAP averages APs across all queries. For the query expansion optimisation WordNet adds "waver", "quiver". These terms are general synonyms of "flutter" that match documents about non-aerospace vibration. Query drift occurs, pulling irrelevant documents that rank above relevant ones. No optimisation ranks document 658 as 2, while query expansion drops to 5. AP gets worse 0.146 -> 0.029, decreasing MAP. Preprocessing wins.

“I used Google Gemini only for Condition 2 surface polish of Task 2 and 3 in WRITEUP.pdf. I used it for clarity and language polishing. I reviewed and revised the edits. No technical ideas, code, analysis, evidence, results, conclusions, report structure, or demonstration content were generated or changed.”